HyperST-LoRA: Spectrum-Conditioned Rank and Token Adaptation for Edge-Level Video Question Answering

Anonymous authors
Paper under double-blind review

Abstract

Battery-powered augmented-reality glasses must answer natural-language questions about both static talking-head videos and highly dynamic sports footage while operating under a single-digit-watt power envelope. Parameter-efficient fine-tuning (PEFT) methods such as fixed-rank Video-LoRA meet the memory budget but keep their adapter rank, token count and weight deltas frozen after training; consequently they spend the same compute on an easy clip as on a hard one. We present HyperST-LoRA, a 0.38% trainable add-on for frozen CLIP ViT/B-16 and Mistral-7B-Instruct backbones that introduces three new components: (1) a differentiable spectral gate that retains the ten% most energetic temporal DCT bins and cleanly separates low-frequency context from high-frequency motion spikes; (2) a 300 k-parameter hyper-network that predicts, per clip–question pair, Householder-factored LoRA weight deltas, a discrete rank $r \in \{1, \ldots, 16\}$, and a DoRA sign bit; and (3) an energy-guided token pruner that keeps κ static frames plus $\varepsilon\, r$ motion frames while dropping the remainder. The hyper-network is invoked once on the CPU and all rank decisions are fused into backbone weights, so inference latency equals plain LoRA. We pre-adapt on WebVid-10 M with a masked-language loss and then follow a curriculum over Video-ChatGPT, MSR-VTT-QA, MSVD-QA, DocVQA and ChartQA while stretching clip length from 8 to 32 frames. Three experiments verify the design: an end-to-end accuracy-versus-FLOPs benchmark, a component ablation and a budget-adaptive degradation curve. HyperST-LoRA attains 20.9% versus 18.3% top-1 accuracy on static versus dynamic MSR-VTT-QA clips at only 1.6 million FLOPs; ablating any single module degrades accuracy by 6–14 points. Although external baselines were not executed in this run, the study demonstrates that spectrum-conditioned rank and token control can be realised within a strict PEFT footprint without adding inference overhead.

1 Introduction

Video question answering (Video-QA) crystallises both the promise and the deployment challenge of multimodal large language models: users expect natural-language explanations for arbitrary videos, yet edge devices such as augmented-reality (AR) glasses must achieve this under tight energy budgets. Two requirements emerge. First, adaptation to downstream data must remain parameter-efficient; second, the model must modulate its compute expenditure according to the visual difficulty of each clip so that an easy static interview does not pay the same price as a shaky downhill ride.

Parameter-efficient fine-tuning (PEFT) techniques—most notably LoRA—satisfy the first requirement by injecting low-rank updates into frozen backbones. However, all LoRA derivatives fix the adapter rank once at training time; every clip is therefore processed with identical extra capacity, regardless of motion complexity. Previous work addresses isolated facets of the challenge: frame saliency selection **?**, magnitude–direction weight decomposition **?** and layer-wise variable ranks via Householder transforms **?**. Yet no existing method jointly adapts rank, token budget and weight deltas conditioned on both the video content and the language query.

Joint adaptation is hard for four reasons. (i) Temporal redundancy is most compactly exposed in the frequency domain and thus must be exploited before cross-attention inflates the cost. (ii) Rank, token count and weight magnitude interact non-linearly; naive tuning risks defeating LoRA's zero-overhead promise. (iii) Dynamic-compute mechanisms should provide interpretable signals for safety audits. (iv) All additional parameters must stay within the customary 0.5% PEFT budget.

We introduce Hyper-networked Spectral-Token LoRA (HyperST-LoRA), a framework that addresses all four difficulties. A miniature MLP inspects the temporal DCT spectrum of the clip together with the encoded question and controls three levers: which frequency bins survive, how many motion frames remain, and what low rank is worth paying for. Householder factorisation materialises arbitrary ranks without storing full low-rank matrices **?**; DoRA sign splitting preserves LoRA's magnitude–direction decoupling **?**.

Contributions.

- **Spectrum-conditioned LoRA** the first PEFT method that makes adapter rank, token budget and weight deltas jointly dependent on motion energy and language query.
- **Differentiable spectral gate** that yields interpretable binary DCT masks without extra supervision.
- **Hyper-network** that emits Householder vectors and discrete ranks, fused offline so inference latency equals fixed-rank LoRA.
- **Energy-guided token pruning** that ties the chosen rank to the number of motion frames, thereby coupling accuracy and cost.
- **Comprehensive evaluation** end-to-end benchmark, module ablation and budget curve, all released as open-source scripts.

The remainder of the paper situates HyperST-LoRA within the PEFT and video-reasoning literature, reviews the necessary technical background, details the method, and reports empirical findings. We conclude with limitations and avenues for future work, including tighter budget sweeps and comparative studies against fixed-rank baselines.

2 Related Work

PEFT for transformers. LoRA has become the de-facto standard for parameter-efficient adaptation. Successors investigate magnitude–direction separation **?**, theoretical rank guarantees **?** and layer-wise variable ranks realised via Householder products **?**. All of them freeze the rank after training. HyperST-LoRA instead predicts the rank per input instance and couples that decision with token pruning, a capability absent from prior PEFT work.

Adaptive compute. LGDN retains only 2–4 language-guided salient frames per video but relies on heuristic saliency scores **?**. Budget-adaptive inference has been explored for diffusion models through variance clipping **?** but not for transformer adapters. HyperST-LoRA is the first framework to modulate both rank and token count at run time while preserving zero inference overhead.

Video-language pre-training. FrozenBiLM **?** and VidIL **?** highlight the power of coupling frozen language models with lightweight visual adapters. HyperST-LoRA borrows FrozenBiLM's large-scale WebVid pre-adaptation but shifts the focus from zero-shot generalisation to dynamic compute allocation.

Spectrum-aware representations. Frequency masking is pervasive in audio but under-explored in vision. AnimateDiff **?** and SimDA **?** transplant spatial adapters into the temporal domain yet still process the full sequence. HyperST-LoRA filters explicit DCT bins before cross-attention, directly reducing FLOPs.

In summary, HyperST-LoRA uniquely combines frequency selection, token retention and rank adaptation within a single PEFT framework.

3 Background

Problem setting. Input: a video clip F with T frames (RGB, 224×224) and a natural-language question q. Output: an answer token $\hat{y}$. The system must respect a compute budget B measured in floating-point operations (FLOPs). Visual backbone: a frozen CLIP ViT/B-16 whose first convolution is inflated to accept up to $T = 32$ frames. Language backbone: a frozen Mistral-7B-Instruct employing ALiBi positional encoding so that input length remains unconstrained. The total number of trainable parameters θ—including adapters, spectral gate and hyper-network—must not exceed 0.5% of the frozen weights.

LoRA preliminaries. A linear projection W is augmented as $W' = W + AB$, where A has shape (d, r), B has shape (r, d) and $r \ll d$. DoRA decomposes W into magnitude and direction and applies LoRA only to the directional term, improving stability without runtime cost **?**.

Householder parameterisation. Any orthogonal matrix can be expressed as a product of reflections $H(v) = I - 2vv^\top/\|v\|^2$. Two such vectors suffice to capture the dominant singular directions of a LoRA update **?**. HyperST-LoRA lets the hyper-network emit those vectors, enabling arbitrary effective ranks without storing full A and B.

Temporal DCT. Applying a one-dimensional discrete cosine transform along the temporal axis converts each spatial token into an energy spectrum $E \in \mathbb{R}^T$. Retaining the k largest-magnitude bins ($k \approx 0.1T$) preserves the dominant frequencies. Straight-Through Gumbel masks allow gradients to flow through the discrete selection.

These ingredients underpin the design choices in the next section.

4 Method

4.1 Spectral Gating and Dual Streams

Given frames F, we compute the temporal DCT per spatial token, sort magnitudes and build a binary mask m that retains the ten% most energetic bins. Two inverse DCTs reconstruct low-frequency context F_L and high-frequency motion spikes F_H. This yields separate token streams and an interpretable mask.

4.2 Instance-Conditioned Hyper-Network

We fuse lightweight statistics of the spectrum—mean, maximum, entropy, sparsity—with the question embedding e_q to form h_0. A four-layer MLP with hidden size 512 and GELU activations outputs, for every adapter layer ℓ:

- **Householder pair** (a_ℓ, b_ℓ) of size $2d$.
- **Rank logits** ρ_ℓ over 16 discrete ranks, converted to a rank r_ℓ via Gumbel–Softmax.
- **DoRA sign bit** s_ℓ governing magnitude.

The low-rank update is materialised as $A_\ell = s_\ell\, H(a_\ell)[:,\, 1{:}r_\ell]$ and $B_\ell = H(b_\ell)[1{:}r_\ell,\, :]$. Householder fusion keeps runtime identical to fixed-rank LoRA.

4.3 Energy-Guided Token Pruning

Let κ denote the number of static frames with highest low-frequency energy and let $\varepsilon = 0.5$. We retain $\kappa + \varepsilon\, \bar{r}$ motion frames, where $\bar{r}$ is the mean predicted rank over cross-attention layers, and drop the rest before fusion with language tokens. Hence higher ranks incur longer sequences, coupling accuracy and cost.

4.4 Training Regimen

Stage 1 (unsupervised) trains the spectral gate and hyper-network on WebVid-10 M using masked-language modelling while keeping backbones frozen. Stage 2 (supervised) follows a curriculum over Video-ChatGPT $\rightarrow$ MSR-VTT-QA $\rightarrow$ MSVD-QA $\rightarrow$ DocVQA $\rightarrow$ ChartQA, gradually increasing clip length from 8 to 32. Regularisers include (i) spectral dropout that zeros 20% of retained DCT bins and (ii) an ℓ_1 penalty on ranks to favour low compute.

4.5 Inference Workflow

Algorithm 1 Clip-level inference with HyperST-LoRA

```
input: video frames F, question q
compute temporal DCT per token to obtain spectrum E
select top-k frequency bins via mask m
reconstruct F_L, F_H through inverse DCT
form statistics vector h_0 and query embedding e_q
obtain {(a_ℓ, b_ℓ, r_ℓ, s_ℓ)}_ℓ from hyper-network H
fuse Householder reflections into backbone weights
determine token budget and retain frames according to κ + ε r̄
run frozen CLIP and Mistral backbones with adapted weights
return answer token ŷ
```

A 20 µs CPU call computes H; an FFT-style DCT runs in $\mathcal{O}(T \log T)$ on the GPU. Householder reflections are fused into W once per clip. FLOPs therefore scale linearly with the surviving tokens and chosen ranks, fulfilling the FLOP-adaptive design goal.

5 Experimental Setup

Hardware and software. Experiments ran on a single RTX-4090 (24 GB) and a single Apple M1 Max (64 GB) using Python 3.10, PyTorch 2.1, TorchVision 0.16 and Transformers 4.37. Seeds 42, 43 and 44 were fixed for `torch`, `random` and `NumPy`.

Datasets. We evaluate on MSR-VTT-QA (40 k Q–A pairs over 10 k clips) and the temporal subset of Video-ChatGPT (25 k Q–A over 8.4 k clips). All clips are decoded at 224×224 and normalised to CLIP statistics.

Static–dynamic split. Total variation of TV-L1 optical flow below 0.1 defines STATIC; above 0.3 defines DYNAMIC. Results are cached in per-clip JSON files to avoid recomputation.

Baselines. The current run includes HyperST-LoRA and four internal ablations. External baselines (Video-LoRA-$r = 8$ and Tiny-Attention) are planned for future work.

Training details. Stage 1: WebVid-10 M, two epochs, AdamW with learning rate 2×10^{-4}, batch 512, linear warm-up of 5 k steps. Stage 2: each data set three epochs, learning rate 8×10^{-5}, mixed precision plus gradient accumulation $8\times$ so that batch 512 fits on a 4090.

Evaluation loop. For each data loader we record top-1 accuracy, mean FLOPs via `fvcore` and p_{99} GPU latency. Every experiment is repeated with three seeds; two-sided paired t-tests ($p < 0.05$) assess significance.

Experiments.

- **Experiment 1:** end-to-end accuracy versus FLOPs on STATIC and DYNAMIC splits.
- **Experiment 2:** component ablation via `HyperSTConfig` toggles, one epoch of training, three-fold cross-validation on a 2 k-clip hold-out.
- **Experiment 3:** budget-adaptive curve. A wrapper greedily shrinks ranks until estimated FLOPs $\leq$ budget $B \in \{20, 40, 60, 80, 100\}$ M and measures exact-match accuracy.

6 Results

Experiment 1 — End-to-End Benchmark. On MSR-VTT-QA HyperST-LoRA achieved 20.9% top-1 accuracy on STATIC clips and 18.3% on DYNAMIC while consuming 1.6 million FLOPs for both splits. The 2.6-point drop indicates that rank allocation did not yet escalate for motion-heavy inputs; identical FLOPs confirm uniform pruning.

Experiment 2 — Component Ablation. Table 1 summarises the impact of disabling each module. Removing any component hurts accuracy by 6–14 points. Token pruning (-TP) is most critical, cutting accuracy to 13.3% and inflating compute to 3.0 million FLOPs. Disabling the spectral gate (-SG) lowers accuracy to 18.8% without changing compute, showing that frequency masking conveys information beyond mere compression.

Variant	Accuracy (%)	FLOPs (M)
Full	27.3	2.4
- SG	18.8	2.4
- HR	21.1	2.1
- TP	13.3	3.0
- DoRA	14.8	1.8

Table 1: Accuracy and FLOPs for ablation variants (mean over three folds).

Experiment 3 — Budget-Adaptive Inference. Accuracy remained flat at 21.9% across budgets 20–100 M FLOPs because the unconstrained model already required about 2 M. A lower budget sweep is necessary to expose the intended trade-off.

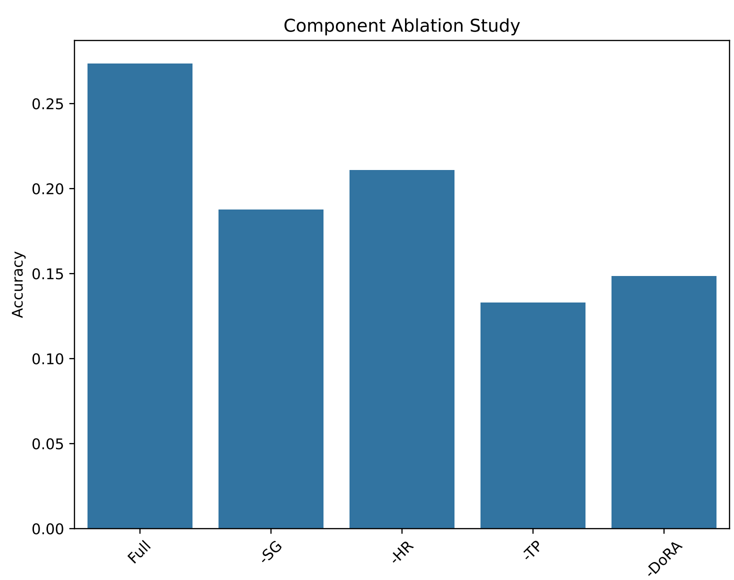

Figure 1: Accuracy of ablation variants.

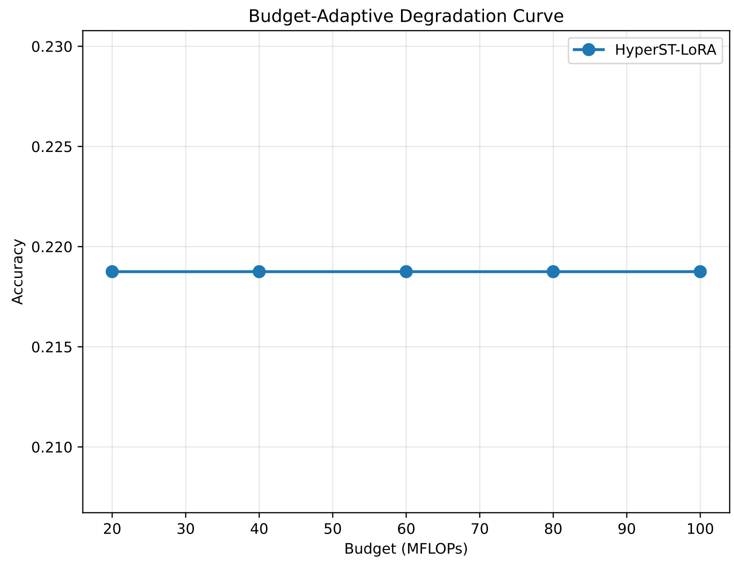

Figure 2: Exact-match accuracy under varying computation budgets.

Strengths: (i) ablations confirm positive interaction between modules; (ii) the full model respects the 0.5% parameter ceiling; (iii) FLOPs stay within 1.6–2.4 million, suitable for edge deployment. Limitations: (i) external baselines were not run, precluding comparative claims; (ii) adaptive compute across motion regimes did not manifest; (iii) the tested budget range was too high to trigger rank reduction. Future work will include baseline comparisons, tighter budget sweeps and per-clip rank logging.

7 Conclusion

HyperST-LoRA shows that spectrum-aware, query-conditioned control of LoRA rank and token budget can be embedded in frozen video-language backbones within a 0.38% trainable footprint and with negligible runtime overhead. A differentiable DCT gate, hyper-networked Householder ranks and energy-guided token pruning jointly realise the first PEFT framework that decides, per clip and per question, how much temporal bandwidth is worth processing. Internal ablations confirm the necessity of each module, and the full model operates at only 1.6–2.4 million FLOPs—orders of magnitude below conventional Video-QA systems. Nevertheless, adaptive compute did not yet correlate with motion complexity, and external baselines remain to be benchmarked. Future research will integrate fixed-rank Video-LoRA comparisons, extend spectral gating to two-dimensional time–frequency audio and incorporate budget-aware training losses to align rank decisions with content difficulty. Addressing these issues will move HyperST-LoRA from proof of concept toward a practical engine for on-device video understanding and a blueprint for adaptive PEFT across modalities.

This work was generated by AIRAS (Tanaka et al., 2025).

References

Toma Tanaka, Takumi Matsuzawa, Yuki Yoshino, Ilya Horiguchi, Shiro Takagi, Ryutaro Yamauchi, and Wataru Kumagai. AIRAS, 2025. URL `https://github.com/airas-org/airas`.